Hidden Breath: Unveiling Dark Oxygen Generation in Abyssal Nodule Ecosystems and the Threat of Deep-Sea Mining

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Abstract

Deep-sea polymetallic nodules function as natural geo-batteries, driving abiotic water splitting and generating molecular oxygen ("dark oxygen") in total absence of sunlight. Specialized microbial communities further augment this oxygen pool via enzymatic dismutation of chlorite and nitric oxide. Together, these processes create oxygenated microhabitats deep on the abyssal plain, sustaining chemosynthetic food webs, modulating carbon remineralization, and altering nutrient cycles. Here, we (1) delineate the electrochemical and microbial pathways responsible for dark O_2 production; (2) quantify global and local fluxes and their dependence on environmental parameters; (3) assess the catastrophic consequences of polymetallic-nodule removal for abyssal oxygen budgets and benthic ecosystems; and (4) propose mitigation technologies, monitoring frameworks, and policy measures to safeguard dark-oxygen sources in the face of deep-sea mining.

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Chapter 1

Introduction

1.1 Concept and Terminology

Below 4000 m, sunlight is absent—yet polymetallic nodules act as natural "geo-batteries," harvesting lithogenic redox energy to drive water-splitting reactions that generate O_2 in complete darkness. Key definitions:

- **Dark oxygen**: Molecular O_2 produced abiotically on metal-oxide surfaces or microbially via dismutation in the abyssal realm.
- **Geo-battery**: A natural electrochemical cell formed by redox gradients on nodule surfaces.
- **Cld**: Chlorite dismutase, enzyme catalyzing $ClO_2{}^- \longrightarrow Cl^- + \frac{1}{2}\,O_2$.
- **Nod**: Nitric-oxide dismutase, enzyme catalyzing $2\,NO \longrightarrow N_2 + O_2$.
- **OER**: Oxygen-evolution reaction, $2\,H_2O \longrightarrow O_2 + 4\,H^+ + 4\,e^-$.

1.2 Historical Overview & Discovery

Early abyssal benthic-chamber measurements recorded net O_2 consumption only. In mid-2024, Sweetman *et al.* observed increases from 185 µmol/L to 819 µmol/L over 48 h atop Clarion–Clipperton Zone nodules, implicating abiotic electrochemistry [1]. Subsequent work confirmed nodule voltages up to 0.95 V and identified microbial dismutases (Cld, Nod) that contribute secondary dark-O_2 flux [3].

1.3 Significance & Broader Impacts

- **Chemosynthetic food webs**: Dark O_2 sustains benthic respiration independent of surface carbon [20].
- **Carbon cycling**: Abyssal fluxes ($1.7\,\mathrm{mmol/(m^2\,day)}$ to $18\,\mathrm{mmol/(m^2\,day)}$) modulate remineralization pathways [1, 2].
- **Policy**: Discovery has spurred calls for moratoria on nodule extraction to prevent localized anoxia [4].

Chapter 2

Geology & Mineralogy of Polymetallic Nodules

2.1 Spatial Distribution & Abundance

2.1.1 Global Hot-Spots

Nodules blanket abyssal plains worldwide; highest densities occur in the Clarion–Clipperton Zone (CCZ), Peru Basin, and Central Indian Ocean Basin [5, 6].

2.1.2 Areal Coverage and Resource Estimates

In the CCZ, nodules cover up to 70 % of the seafloor with standing stocks of $10\,\mathrm{kg/m^2}$ to $15\,\mathrm{kg/m^2}$. Estimated total dry nodule reserves exceed 21×10^9 t, containing 5.95×10^9 t Mn, 0.27×10^9 t Ni, 0.23×10^9 t Cu, and 0.05×10^9 t Co [5, 6].

2.2 Nodule Composition & Structure

2.2.1 Mineral Phases

Dominant manganese oxides:

- Vernadite (δ-MnO_2).
- Todorokite (3×3 tunnel structure).
- Birnessite, Asbolan (minor).
- Intergrown Fe-oxyhydroxides (goethite, ferrihydrite).

Diagenetic nodules exhibit Mn/Fe ratios > 10 and trace Ni (1.3 %), Cu (1.1 %), Co (0.2 %) [6].

2.2.2 Trace Metal Content

- Diagenetic nodules: high Mn/Fe, Ni up to 1.3 %, Cu up to 1.1 % [6].
- Hydrogenetic nodules: enriched in Co, REEs, Ti, Zr, Nb; Mn/Fe ≤ 5 [6].

2.2.3 Physical Characteristics

- Porosity 25 % to 61 %; dry density 1.0 g/cm^3 to 2.4 g/cm^3 [7].
- Specific surface area 100 m^2/g to 420 m^2/g [8].
- Extensive microfracture networks facilitate fluid transport [7].

2.3 Electrochemical Properties

2.3.1 Conductivity in MnO_x Lattices

Layered oxides (birnessite) are semiconducting (band gap 1.7 eV) with enhanced conductivity via Mn^{3+}/Mn^{4+} vacancy ordering [9].

2.3.2 Natural Redox Gradients and Geo-battery Potentials

In situ voltage differences up to 0.95 V develop between anodic (Fe^{2+}, Mn^{2+} oxidation) and cathodic microdomains, driving OER [1].

2.3.3 Surface Catalytic Sites

Mn-oxide clusters catalyze OER with overpotentials 300 mV to 400 mV at pH 7.5 and 2 °C, proceeding via a Mars–van Krevelen mechanism [9].

Chapter 3

Abiotic Electrochemical Oxygen Generation

3.1 Geobattery Model

3.1.1 Anodic vs. Cathodic Zones

Polymetallic nodules act as natural galvanic cells. In anodic microdomains, Fe^{2+} and Mn^{2+} are oxidized, releasing electrons:

$$M^{2+} \longrightarrow M^{3+} + e^-$$

These electrons traverse the MnO_x lattice to cathodic sites, where they drive water-splitting reactions [1, 11].

3.1.2 Lithogenic Electron Donors

The primary electron donors are dissolved Fe^{2+} and Mn^{2+} from the nodule matrix. Under abyssal conditions, dissolution currents of $0.1\,\mu A/cm^2$ to $1\,\mu A/cm^2$ sustain continuous O_2 evolution [21].

3.2 Water-Splitting Reactions

3.2.1 Half-Reactions and Standard Potentials

At the cathode, water oxidation and proton reduction proceed as:

$$2\,H_2O \longrightarrow O_2 + 4\,H^+ + 4\,e^- \quad (E^\circ = +1.23\,V)$$

$$4\,H^+ + 4\,e^- \longrightarrow 2\,H_2 \quad (E^\circ = 0.00\,V)$$

In situ cell potentials up to 0.95 V exceed the threshold for O_2 formation, albeit kinetically limited by overpotential [17].

3.2.2 Kinetic Parameters

The oxygen-evolution reaction (OER) on MnO_x follows a Butler–Volmer form with:

$$\text{Tafel slope } b = 65\,\text{mV/decade}, \quad j_0 = 1 \times 10^{-6}\,A/cm^2.$$

These parameters reflect deep-sea conditions and mixed-valence manganese oxides [10, 11].

3.2.3 Pressure–Temperature Corrections

High pressure (400 bar) and low temperature (2 °C) shift the standard potential by 15 mV and reduce reaction rates by approximately 30% [2].

3.3 Lattice-Oxygen Mechanisms

3.3.1 Mars–van Krevelen Cycles

Lattice O^{2-} participates directly in O_2 formation, creating oxygen vacancies that water subsequently replenishes, as evidenced by isotope exchange studies [12].

3.3.2 ^{18}O Isotopic Tracing

Experiments in ^{18}O-enriched seawater show that roughly 25% of evolved O_2 originates from lattice oxygen, demonstrating a mixed mechanism of surface and lattice oxygen evolution [12, 8].

3.3.3 Electrochemical Impedance Spectroscopy

EIS on nodule fragments reveals charge-transfer resistances of $100\,\Omega\,\mathrm{cm}^2$ to $300\,\Omega\,\mathrm{cm}^2$ and Warburg diffusion elements, indicating mass-transport limitations within the porous structure [11].

3.4 Quantification Techniques

3.4.1 In-Situ Voltammetric Microelectrodes

Cyclic voltammetry in benthic chambers measures OER onset potentials and current densities directly on nodule surfaces, enabling real-time flux estimation [21].

3.4.2 Microsensor Array Mapping

Clark-type O_2 microelectrodes (10 µm tip) map concentration gradients; fluxes are calculated via Fick's law:

$$J = -D_{\mathrm{eff}} \frac{\partial[\mathrm{O}_2]}{\partial x}$$

with $D_{\mathrm{eff}} \approx 1 \times 10^{-5}\,\mathrm{cm}^2/\mathrm{s}$ [19].

3.4.3 Normoxic vs. Hyperoxic Baselines

Background abyssal O_2 is 180 µmol/L to 220 µmol/L; values exceeding 400 µmol/L near nodules confirm active production [1].

Chapter 4

Microbial Pathways to Dark Oxygen

4.1 Key Biochemical Reactions

Microbial dismutation pathways produce O_2 in darkness:

Chlorite dismutation: $\mathrm{ClO_2^-} \longrightarrow \mathrm{Cl^-} + \frac{1}{2}\,\mathrm{O_2}$ (Chlorite Dismutas

Nitric-oxide dismutation: $2\,\mathrm{NO} \longrightarrow \mathrm{N_2} + \mathrm{O_2}$ (Nitric-oxide Dismuta

Methanobactin-mediated water splitting: $\mathrm{H_2O} \longrightarrow \mathrm{O_2} + 2\,\mathrm{H^+} + 2\,\mathrm{e^-}$ (obser

4.2 Community Structure & Ecology

16S rRNA surveys show Cld- and Nod-encoding taxa comprise 10 % to 30 % of nodule-associated communities. *Dechloromonas* ASVs can reach 25 %, while NC10 phylum members contribute 5 % to 10 % [13, 14].

4.3 Enzyme Kinetics & Thermodynamics

Chlorite dismutase exhibits $k_{\mathrm{cat}} = 1 \times 10^3/\mathrm{s}$, $K_M = 10\,\mu\mathrm{M}$, $E_a = 45\,\mathrm{kJ/mol}$ and $Q_{10} = 1.2$, while nitric-oxide dismutase shows $k_{\mathrm{cat}} = 1 \times 10^2/\mathrm{s}$, $K_M =$

1 µM, with similar temperature dependence; both remain active under abyssal P–T conditions [15, 2].

4.4 Coupling of Abiotic & Biotic Processes

Abiotic OER byproducts (H_2, H_2O_2) fuel microbial metabolisms. Biofilm formation enhances effective electrode area by $\approx 30\%$, boosting overall O_2 fluxes in microcosms [3, 20].

Chapter 5

Reaction Kinetics, Thermodynamics & Modeling

5.1 Rate-Law Formulations

5.1.1 Abiotic OER: Butler–Volmer

$$j = j_0 \Big[\exp\big(\tfrac{\alpha n F}{RT}\eta\big) - \exp\big(-\tfrac{(1-\alpha) n F}{RT}\eta\big) \Big],$$

with $n = 4$, $\alpha = 0.5$, $j_0 = 1 \times 10^{-6}\,\mathrm{A/cm^2}$ [11].

5.1.2 Enzymatic Kinetics: Michaelis–Menten & Monod

$$v_{\mathrm{Cld}} = \frac{k_{\mathrm{cat,Cld}}[\mathrm{ClO_2^-}]}{K_{M,\mathrm{Cld}} + [\mathrm{ClO_2^-}]}, \quad \mu = \mu_{\max} \frac{[S]}{K_S + [S]},$$

with Monod parameters for nitric-oxide dismutase from Ettwig *et al.* [16, 15].

5.1.3 Coupled Reaction-Network

$$\frac{d[\mathrm{O_2}]}{dt} = R_{\mathrm{abiotic}}([\mathrm{H^+}], \eta) + R_{\mathrm{biotic}}([S]) - R_{\mathrm{consumption}},$$

incorporating respiration sinks from Glud *et al.* [20].

5.2 Thermodynamic Constraints

5.2.1 Gibbs Free Energy

$$\Delta G = \Delta G^{\circ} + RT \ln\Big(\frac{[H^+]^4}{p_{\mathrm{O_2}}}\Big),$$

shifted by 15 mV under abyssal P–T conditions [2].

5.2.2 Activity Corrections

Ionic strength ($I \approx 0.7\,\mathrm{M}$) modifies activity coefficients via extended Debye–Hückel theory [17].

5.2.3 Mn Redox Speciation

Buffers of $\mathrm{Mn^{2+}}/\mathrm{Mn^{3+}}/\mathrm{Mn^{4+}}$ constrain redox potentials on the Eh–pH diagram, guiding coupled geochemical–biological models [18].

5.3 Reactive Transport Simulations

5.3.1 One-Dimensional Pore-Water Models

$$\frac{\partial C}{\partial t} = D_{\mathrm{eff}} \frac{\partial^2 C}{\partial x^2} + R(C),$$

where $D_{\mathrm{eff}} = 1 \times 10^{-5}\,\mathrm{cm^2/s}$ and reaction terms follow Boudreau [19].

5.3.2 Two-Dimensional Plume Dispersion

Advection–diffusion modeling predicts $\mathrm{O_2}$ halo extents driven by bottom currents, parameterized after Glud *et al.* [21].

5.3.3 Sensitivity Analyses

Parameter sweeps identify critical thresholds for maintaining $[O_2] > 5\,\mu\text{M}$, informing collapse models in subsequent chapters.

Chapter 6

Mass-Balance & Ecosystem Fluxes

6.1 Global Dark-O_2 Budget

6.1.1 Extrapolation from CCZ

Flux measurements on Clarion–Clipperton Zone (CCZ) nodular fields yield rates of $1.7\,\text{mmol}/(\text{m}^2\,\text{day})$ to $18\,\text{mmol}/(\text{m}^2\,\text{day})$. Over the CCZ area of $4.5 \times 10^6\,\text{km}^2$, this implies gross O_2 generation of 7.7 Tg/year to 82 Tg/year, comparable to abyssal respiration sinks of 100 Tg/year to 200 Tg/year [1, 20].

6.1.2 Comparison with Other Sources

Hydrothermal-vent abiotic O_2 production is negligible (<0.1 Tg/year), while chemoautotrophic dark-ocean processes add 5 Tg/year. Thus, polymetallic-nodule dark O_2 may account for 5 % to 40 % of non-photosynthetic abyssal-plain oxygen inputs [20, 21].

6.2 Local Flux Measurements

6.2.1 Benthic-Chamber Experiments

Closed-chamber incubations directly over nodule beds record O_2 accumulation of 5 µmol/(L h) to 15 µmol/(L h), equivalent to 2 mmol/(m^2 day) to 10 mmol/(m^2 day) [1].

6.2.2 Eddy-Covariance O_2 Flux Towers

Turbulent O_2 fluxes measured by eddy-covariance on benthic landers resolve high-frequency exchanges of 1 mmol/(m^2 day) to 8 mmol/(m^2 day), agreeing within 20 % of chamber data [21].

6.3 Downstream Effects

6.3.1 O_2 Diffusion into Sediment Porewaters

Reactive-transport simulations show O_2 produced at the nodule interface diffuses via

$$\frac{\partial C}{\partial t} = D_{\text{eff}} \frac{\partial^2 C}{\partial x^2} + R(C),$$

reaching 1 cm to 3 cm depth, sustaining aerobic diagenesis [19, 18].

6.3.2 Impact on Benthic Nutrient Cycling

Dark O_2 enables aerobic nitrification and sulfur oxidation, increasing nitrification rates by 30 % to 50 % and modulating ${NH_4}^+$ and HS^- profiles [20].

6.3.3 Modulation of Carbon Burial Rates

Enhanced oxic remineralization reduces organic-carbon burial efficiency by up to 10 % in nodule-rich areas, affecting long-term carbon sequestration [1].

Chapter 7

Impact of Polymetallic-Nodule Removal

7.1 Mining Scenarios & Spatial Footprints

7.1.1 Collector Technologies

Bucket-wheel and hydraulic suction systems disturb 100 m to 200 m swaths at 1 kn to 3 kn, reworking up to 5 km^2/year per vessel [5].

7.1.2 Plume Generation and Sediment Resuspension

Collectors generate benthic and bentho-pelagic plumes that redeposit 1 cm to 5 cm of fine sediment on adjacent nodules, smothering O_2-producing biofilms [6, 7].

7.2 Loss of Electrochemical Substrate

7.2.1 Cessation of Geo-battery Water Splitting

Nodule removal eliminates mineral electrodes; models predict >90 % drop in O_2 production within 24 h [1].

7.2.2 O_2 Re-equilibration Timescales

Without production, abyssal waters revert to 200 µmol/L on 7 day to 14 day timescales via vertical diffusion ($D_{\text{eff}} \approx 1 \times 10^{-5}\,\text{cm}^2/\text{s}$) [19].

7.3 Microbial Community Disruption

7.3.1 Habitat Loss vs. Recolonization

Microbial recolonization occurs over decades; cleared areas remain anoxic and oligotrophic for >5 year (DISCOL, JET data) [5].

7.3.2 Gene-Pool Impacts

Sediment metagenomics show >80 % loss of chlorite and nitric-oxide dismutase genes within 100 m of disturbance [13].

7.3.3 Redox Gradient Flattening

Porewater Eh shifts from +200 mV to -100 mV in the top 5 cm, eliminating aerobic niches [20].

7.4 Modeling Collapse Thresholds

7.4.1 O_2 Tipping Points

Bifurcation analysis identifies a critical nodule-cover fraction ($\approx 10\,\%$) below which O_2 cannot self-sustain, leading to abrupt anoxia [1].

7.4.2 Spatially Explicit Ecosystem Collapse

Agent-based models predict >60 % megafaunal biomass loss within 2 year under continuous mining pressure [5].

7.4.3 Synergy with Climate-Driven Hypoxia

Bottom-water warming (+1 °C) and deoxygenation lower collapse thresholds by 15 % to 20 % [2].

Chapter 8

Ecosystem Responses & Case Studies

8.1 Bentho-Pelagic Coupling

8.1.1 Macrofaunal Impacts

Nodule fields host diverse macrofauna; loss of O_2 microhabitats reduces abundance by 40 % to 70 % [6, 20].

8.1.2 Fish & Cephalopod Mortality

Mobile megafauna suffer >50 % mortality within 500 m of disturbance due to prolonged hypoxia [21].

8.2 Experimental De-Noduling Trials

8.2.1 DISCOL & JET Sites

Artificial clearances over 10 m^2 plots show >80 % benthic biomass decline and persistent anoxia for >30 year [5].

8.2.2 Mesocosm Studies

Simulated removal trials yield complete microbial mat collapse and 70 % decline in meiofaunal density within months [13].

8.3 Trophic Web Disruption

8.3.1 Loss of Chemosynthetic Symbionts

Siboglinid tubeworms lose 90% of symbionts post-clearance, leading to host mortality [20].

8.3.2 Scavenger Cascades

Reduced detritivore activity lowers scavenger abundance by 50 % and slows organic-matter turnover [21].

8.4 Comparative Case Study

8.4.1 Impacted vs. Pristine Sites

Metric	Pristine	Impacted	Change
O_2 Flux (mmol/(m^2 day))	12.5	3.1	-75%
Macrofaunal Diversity (Shannon)	2.8	1.1	-60%
Carbon Remineralization (mmol/(m^2 day))	8.4	5.0	-40%

Table 8.1: Ecosystem metrics: Pristine vs. nodule-removed sites [5, 1].

Chapter 9

Mitigation & Management Strategies

9.1 In-Situ Electrochemical Preservation

9.1.1 Artificial Cathode Arrays

Deploy inert cathode arrays (e.g. graphite or titanium plates) galvanically linked to residual mineral fragments. Laboratory microcosms demonstrate sustained water-splitting at 1 mmol/(m^2 day) to 5 mmol/(m^2 day) [3].

9.1.2 Slow-Release Mineral Amendments

Disperse MnO_2–pellet "seabed batteries" that leach redox-active solids over 1 year to 2 year, buffering O_2 production against episodic disturbances [8].

9.2 Regulatory Frameworks

9.2.1 ISA Management Plans

ISA draft plans require monitoring benthic O_2 flux as a KPI and suspend mining if fluxes drop >50 % below baseline [5].

9.2.2 Precautionary Principle

Under UNCLOS, halt activities when "serious harm" is likely—quantified here by dark-O_2 metrics as suspend triggers [4].

9.3 Adaptive Monitoring & Early Warning

9.3.1 Real-Time O_2 Telemetry

Cabled observatories with optode arrays deliver 1 Hz O_2 data, detecting >10 % declines within hours [21].

9.3.2 eDNA Surveys

Metabarcoding of sediment and nodule biofilms tracks Cld/Nod gene-abundance shifts; deviations >50 % over 30 days signal stress [13].

9.3.3 Decision Rules

Suspend mining when both:

1. O_2 flux $< 5\,\mathrm{mmol/(m^2\,day)}$, and

2. Cld/Nod gene loss $> 50\,\%$ over 30 days.

Chapter 10

Conclusions & Future Directions

10.1 Synthesis of Abiotic and Biotic Interplay

Polymetallic nodules act as geo-batteries driving abiotic O_2 evolution, supplemented by microbial dismutation. Together they sustain chemosynthetic benthic ecosystems and modulate carbon and nutrient cycles.

10.2 Knowledge Gaps & Research Priorities

1. **High-Pressure Electrochemistry:** In situ voltammetry at 400 atm.
2. **Deep Metagenomics:** Novel dismutase discovery and regulation.
3. **Coupled Ocean–Climate Models:** Incorporate dark-O_2 fluxes.
4. **Eco-Engineering Trials:** Field tests of geo-cathodes and mineral amendments.

10.3 Broader Implications

- **Astrobiology:** Guides search for abiotic O_2 on ocean worlds.
- **Conservation:** Justifies precautionary deep-sea mining policies.
- **Philosophy:** Expands understanding of non-photosynthetic life support.

Glossary of Terms & Abbreviations

Geo-battery Natural electrochemical cell formed by redox-active polymetallic nodules, driving the oxygen-evolution reaction.

Cld (Chlorite Dismutase) Enzyme catalyzing $ClO_2^- \longrightarrow Cl^- + \frac{1}{2}\,O_2$.

Nod (Nitric-oxide Dismutase) Enzyme catalyzing $2\,NO \longrightarrow N_2 + O_2$.

OER (Oxygen-Evolution Reaction) Electrochemical half-reaction $2\,H_2O \longrightarrow O_2 + 4\,H^+ + 4\,e^-$.

Mars–van Krevelen Cycle Mechanism in which lattice oxygen in metal oxides participates directly in O_2 formation, creating and replenishing oxygen vacancies.

Tafel slope Electrocatalytic parameter (in mV/decade) that quantifies the overpotential dependence of reaction rate.

D_{eff} **(Effective Diffusivity)** Apparent diffusion coefficient in sediment porewaters, accounting for porosity and tortuosity.

Process / Parameter	**Symbol**	**Value (Units)**	**Reference**
OER Tafel slope	b	65–75 mV dec^{-1}	[10]
OER exchange current density	j_0	1×10^{-6}–1×10^{-5} A cm^{-2}	[11]
OER charge transfer coefficient	α	0.50	[11]
Cld Michaelis constant	$K_{M,\mathrm{Cld}}$	10 µM	[14]
Cld turnover number	$k_{\mathrm{cat,Cld}}$	1×10^3 s^{-1}	[14]
Nod Michaelis constant	$K_{M,\mathrm{Nod}}$	1 µM (approx.)	[15]
Nod turnover number	$k_{\mathrm{cat,Nod}}$	1×10^2 s^{-1}	[15]
Effective diffusion coefficient	D_{eff}	1×10^{-5} cm^2 s^{-1}	[19]
Activation energy (Cld)	E_a	45 kJ mol^{-1}	[14]
Pressure correction to standard potential	ΔE°	+15 mV at 400 atm	[2]
Enzymatic Q_{10}	Q_{10}	1.2	[15]

Table A.1: Key kinetic and thermodynamic parameters used in the reaction and transport models.

Appendix A

Kinetic Thermodynamic Parameters & Model Equations

Model Equations

$$j = j_0 \Big[\exp\big(\tfrac{\alpha\, n\, F}{RT}\,\eta\big) - \exp\big(-\tfrac{(1-\alpha)\, n\, F}{RT}\,\eta\big)\Big], \tag{A.1}$$

$$v_{\mathrm{Cld}} = \frac{k_{\mathrm{cat,Cld}}\,[\mathrm{ClO_2^-}]}{K_{M,\mathrm{Cld}} + [\mathrm{ClO_2^-}]}, \tag{A.2}$$

$$v_{\mathrm{Nod}} = \frac{k_{\mathrm{cat,Nod}}\,[\mathrm{NO}]}{K_{M,\mathrm{Nod}} + [\mathrm{NO}]}, \tag{A.3}$$

$$\frac{d[\mathrm{O_2}]}{dt} = R_{\mathrm{abiotic}}([\mathrm{H^+}], \eta) + R_{\mathrm{biotic}}([S]) - R_{\mathrm{consumption}}, \tag{A.4}$$

$$\frac{\partial C}{\partial t} = D_{\mathrm{eff}}\,\frac{\partial^2 C}{\partial x^2} + R(C). \tag{A.5}$$

Explanation of Model Equations

- **Butler–Volmer equation for current density j:** Relates the net electrochemical oxygen-evolution current to the overpotential η at the nodule "electrode." The two exponential terms represent the anodic (oxygen-producing) and cathodic (hydrogen-producing) branches.

- **Michaelis–Menten rates v_{Cld} and v_{Nod}:** Describe enzyme-catalyzed oxygen production by chlorite dismutase and nitric-oxide dismutase as functions of substrate concentration, capturing turnover rates and substrate affinities measured in deep-sea isolates.

- **Oxygen mass-balance ($d[O_2]/dt$):** Sums abiotic generation (function of proton concentration and overpotential), biotic generation (enzymatic rates), and all consumption terms (respiration, chemical sinks) to yield the net change in dissolved O_2.

- **Reactive-transport equation ($\partial C/\partial t$):** Models diffusion of dissolved species (e.g. O_2) through porous sediment (with effective diffusion coefficient D_{eff}) and local production/consumption reactions $R(C)$, reproducing measured pore-water concentration profiles.

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